Rodents

PHYLUM	Chordata
CLASS	Mammalia
ORDER	Rodentia
FAMILIES	34
SPECIES	2,478

CLASSIFICATION NOTE

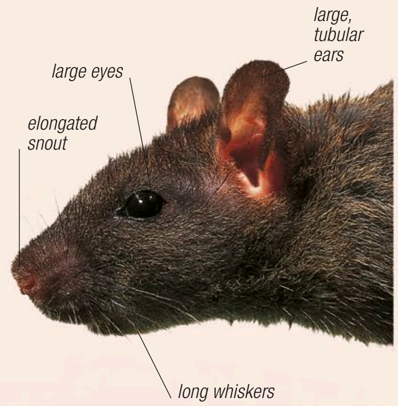
Some mammalogists prefer to divide the order Rodentia into 2 suborders: (Sciurognathi and Hystricognathi). Others advocate a division into 5 suborders, which are defined by jaw musculature: Sciuromorpha (squirrel-like rodents), Castorimorpha (beaverlike rodents), Myomorpha (mouselike rodents), Histricomorpha (cavylike rodents), and Anomaluromorpha (springhare). For greater ease of reference, the latter division is used here.

Squirrel-like rodents see pp.118–21
Beaverlike rodents see pp.121–2
Mouselike rodents see pp.123–8
Cavylike rodents see pp.129–31
Springhares see p.132

Representing over 40 percent of all mammal species, rodents form a successful and highly adaptable order. They are found worldwide (except Antarctica) in almost every habitat: lemmings, for example, favor the cold climate of the arctic tundra, while gundis prefer the heat of African desert regions. Despite the variety of lifestyles and habitats exhibited by members of this order, there are many common characteristics: most rodents are small quadrupeds with a long tail, clawed feet, long whiskers, and teeth (especially the long incisors) and jaws specialized for gnawing. Although generally terrestrial, some species are arboreal (such as tree squirrels), burrowing (mole-rats, for instance, live almost wholly underground), or semiaquatic (such as beavers). Some species, such as the woodchuck, are solitary, but most are highly social and form large communities.

Senses

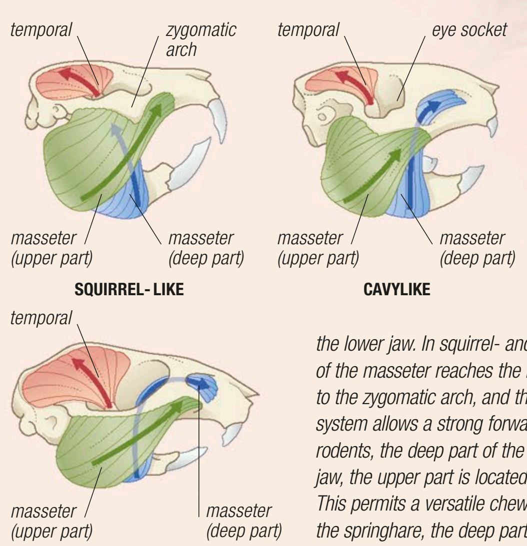
Most rodents enjoy acute senses of smell and hearing, which, in combination with their long and numerous touch-sensitive whiskers, provide them with a heightened awareness of their surroundings. Nocturnal species have larger eyes than diurnal species, to maximize the amount of light received by the retina (the greater the amount of light, the brighter and clearer the image). Rodents communicate by smell (odors are secreted from scent glands on the body) and by an extensive range of vocalizations.



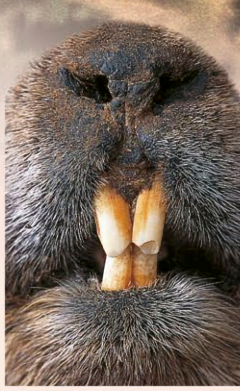
HIGHLY TUNED
Well-developed sense organs are present in most rodents and may contribute to the adaptability of species, such as this brown rat. The large eyes and ears, elongated snout, and long whiskers are typical of many rodents.

Anatomy

While the anatomy of rodents is more uniform than that of most other orders of mammals, some characteristics, such as a compact body and a long tail and whiskers, are shared by many species. The front foot usually has 5 digits (although the thumb may be vestigial or absent), the back foot has 3–5 digits, and the method of locomotion is generally plantigrade. Different species use their tail to perform distinct functions: beavers have a flat, wide tail that is used for steering when swimming, while the Eurasian harvest mouse uses its prehensile tail when climbing in long grass. In some species, part of the tail skin, or the tail itself, will break off if caught, enabling the animal to escape. Because rodent anatomy is more generalized than that of other mammals, they can adapt easily and are able to thrive in many different habitats.



JAW MUSCLES
Rodents have an enlarged chewing muscle (the masseter), which permits both a vertical and a back and forth motion of the lower jaw. In squirrel- and beaverlike rodents, the upper part of the masseter reaches the back of the skull, the deep part extends to the zygomatic arch, and the temporal muscle is small. This system allows a strong forward motion when biting. In mouselike rodents, the deep part of the masseter extends onto the upper jaw, the upper part is located forward, and the temporal is large. This permits a versatile chewing action. In cavylike rodents and the springhare, the deep part of the masseter extends in front of the eye, and the temporal is small. This gives a strong forward bite.



LARGE INCISORS
The 4 huge incisors (seen here in a marmot) distinguish rodents from most other mammal orders. These teeth are long, curved, and grow continually. Only the front surface of these teeth has enamel, however—the back surface consists of softer dentine, which is eroded by constant gnawing to ensure that the teeth remain sharp.